

Ex.No. 12:

NB-IoT vs LTE-M Throughput Simulation Date

Objective 1 :

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

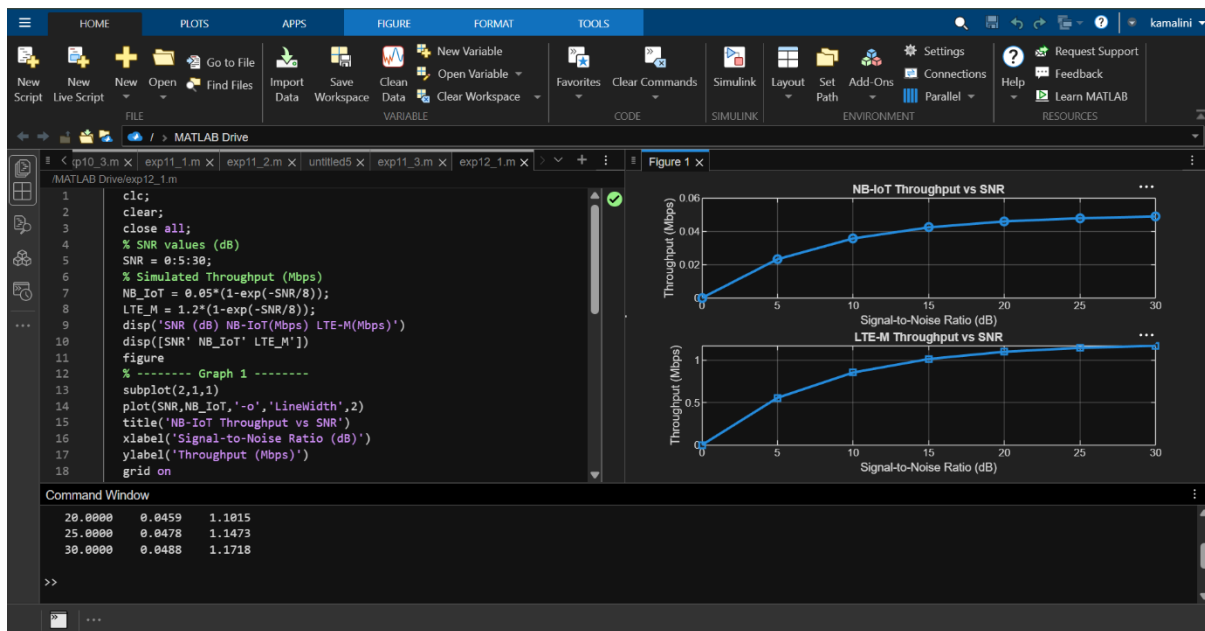
CODE :

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));
disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
```

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on



Objective 2 :

To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

CODE :

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

LTE_M = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

figure

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR,NB_IoT,'-o','LineWidth',2)
```

```
title('NB-IoT Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

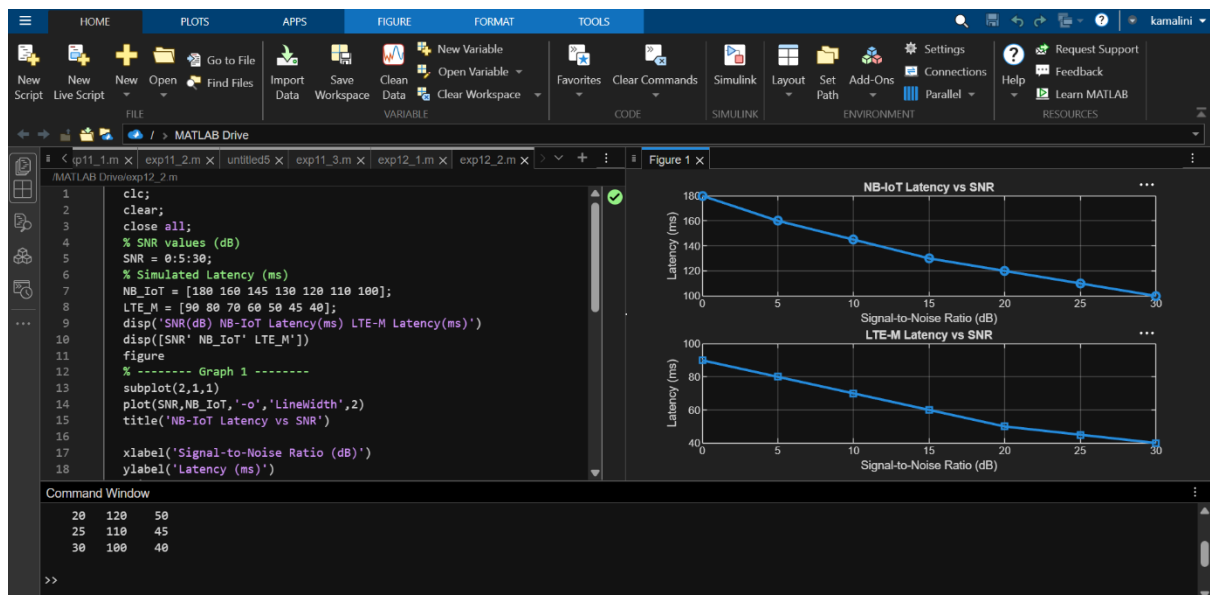
```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

```
title('LTE-M Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```



Objective 3 :

To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

CODE :

```

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB_IoT = [82 86 89 92 95 97 99];

LTE_M = [88 91 94 96 98 99 100];

disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')

disp([SNR' NB_IoT' LTE_M'])

```

```

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

```

